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INTERLEAF

PROCESS AND AUTOMATION

Valmet Standard for SHOTTONPM3-Project, Process and Automation

	PROJECT INFORMATION
PROJECT NAME	Shotton Mill Ltd.
PROJECT CODE *)	SHOTTONPM3
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NOTE	

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*) Max. 16 characters (ProjKey-Common Keywords)

REVISIONS

0 Description of 1st revision

TABLE OF CONTENTS

1 PROJECT SPECIFIC INFORMATION

Content	Responsible	Rev.
1.1 Voltage levels and signal levels	Supplier and purchaser	5
1.2 AC-Motors, frame types and circuit diagrams	Supplier and purchaser	1

2 Valmet STANDARD FOR ENGINEERING AND MANUFACTURING

Content	Rev.
2.1 Positioning, automation and process	7
2.2 AC-Motor, positioning and control principles	2
2.3 Positioning, QCS	0
2.4 Positioning, lubrication and condition monitoring	1
2.5 Terminal blocks and Internal wiring	4
2.6 Colors of push buttons and pilot lights	0
2.7 Cable numbers and principles	8
2.8 Field Cable Types	2
2.9 Hydraulic, lubrication and pneumatic piping	1
2.10 Measuring units	3
2.11 I/O Interface Methods	2
2.12 Grounding principles	1
2.13 Process Engineering Instructions	1
2.14 General control principles and definitions of process automation	0

1.1 Voltage levels and signal levels

1.1.1 Supply voltages

INFORMATION RECEIVED FROM SHOTTON MILL ON MAY 30, 2022 UPDATED LATER

Application	Description	Voltage/frequency
AC-motor voltages	Power and motor voltage	
	(motors > 800 kW)	11 kV, 50 Hz
	(motors ≤ 800 kW)	690 VAC, 50 Hz
	(motors < 90 kW)	400 VAC, 50 Hz
	Frequency controlled motors	
	(motors ≥ 90 kW)	690 VAC, 50 Hz
	(motors < 90 kW)	400 VAC, 50 Hz
	MCC starter control voltage	110 VAC, 50 Hz
	MCS/DCS - MCC signal voltage	24 VDC
	(Analog signals:Galvanic isolation at MCC end)	
	Control voltage for contactors and power supply for instrumentation is	
		230 VAC 110 VAC , 50 Hz
> 1000V motor voltages	Starter control voltage	VDC
	MCS/DCS - MCC signal voltage	VDC

	(Analog signals:Galvanic isolation at MCC end)	
Sectional drives	Motor voltage	690 VAC, 50 Hz
Lighting / field supplies	Supply voltage	400 3-ph / 230 VAC, 50 Hz W/E and inside hood- 110 VAC, 50 Hz
Automation system	Supply voltage	230 VAC, 50 Hz
	UPS supply voltage from customer	110 and 230 and 400 VAC, 50 Hz, 30 kVA/ 30 min
AC-voltage system grounding method	TN-S (separate neutral and grounded wires)	400 VAC TN-S
	TN-C (neutral and grounded in the same wire, PEN-wire-)	690 VAC TN-C
	IT (system isolated from ground)	690V,50 Hz, resistance grounded neutral point (IT system)
	TT (one point of circuit grounded)	

1.1.2 Control Voltages

The DC voltage level for the DNA interface boards is 28 VDC and PLC interface boards 24 VDC.

Connections to other voltage levels and high power devices (> 3 W) are made by using input and output interface units (relays, opto-couplers).

Valmet uses following voltage levels:

Application	
Control switches and pushbuttons	24 VDC
Pilot lights	24 VDC
Proximity switches	24 VDC
Pneumatic solenoid valves	24 VDC
Hydraulic solenoid valves	24 VDC
Interlocking signals to other control systems - with optical isolator or - pot. free relay contact or - serial interface (optional)	24 VDC
MCC-MCS signals	24 VDC
Analog input and output signals - Field equipment's without Hart connections - Active transmitters at field have to have galvanic isolation in transmitter end	4 - 20 mA

1.2 AC-Motors, frame type and circuit diagram

1.2.1 AC-Motor Frame Type

Motor frames	IEC	

1.2.2 Control Type

Connection method:

MCC - MCS/DCS	In use
Relay control	
Profinet	x

~~Profibus units in MCC:~~

Type of Profibus DP units -x = In use (x) = In use when needed	In use	Type
Basic module	x	Simocode Pro V
Extension modules		
(Power, I/O, etc...)		
ProfiSafe module (For PLC or PLd safety class)	(x)	DM-F Local
		DM-F Profisafe

1.2.3 Motor Circuit Diagrams

Sample drawings needed from Customer / to be sent to customer

- Normal
- Variable speed
- Reversing
- Two speed
- Forward / reversing

- With brake
- Heavy start-up
- Heating

2.1 Positioning, automation and process

Valmet uses the system of circuits (functional groups) where each machine function in machine section has its own number designated for its use. That's why Valmet can deliver circuit drawings quite fast and have stable quality for its deliveries.

The quality is based on reference solutions of circuits. The solutions are tested and approved. Position codes, which are based on circuit numbers, are used in documents of different application areas. The software of machines is also based on the circuit numbering system. Changes in principles of circuits will cause changes in software structures.

The machines are divided into functional parts, numbered circuits. All the devices (electric, pneumatic and hydraulic) in the circuit are positioned using the same circuit number.

Positioning system based following standards,

- IEC81346-1, Industrial systems, installations and equipment and industrial products. Structuring principles and reference designations.

Part 1: Basic rules

- IEC81346-2, Industrial systems, installations and equipment and industrial products. Structuring principles and reference designations.

Part 2: Classification of objects and codes for classes

- IEC62424, Representation of process control engineering – Requests in P&I diagrams and data exchange between P&ID tools and PCE-CAE tools

- ISO14617-6, Graphical symbols for diagrams. Part 6: Measurement and control functions

2.1.1 Cross Direction Actuators

Cross direction actuators	Numbering starting from tending side	TS -> DS 1,2,3...
Cross direction measurements	CD-profile shown on ValmetDNA display	TS Left DS Right
Numbering of slitters at winder	Numbering starting from left or right when looking towards the web direction	Left -> Right 1,2,3...

2.1.2 Location Codes

MM-SSAAA-CCC.DD

M Plant Area code
 SS Sub- Process code
 AAA Letter code (2-3 characters)
 CCC Serial number within process area (1-3 characters)
 DCS and MCS, consecutive numbers will be used as follows:
 001 ... 499 MCS (EXCEPTION ! Winder MCS use 001 ...
 999 area)
 500 ... 999 DCS
 DD Field number, if required (1-2 characters)

Letter codes:

Letter code (AAA)	Description
CB	Control box
CD	Control desk
CP	Control Panel, Operator Terminal etc
VD	Control Voltage Distribution Panel
CR	PLC cabinet, control cabinet, Cross Connection Rack, Control Rack (including. DCS I/O)
RR	Relay rack
PC	Input/Output Rack
FB	Field box
JB	Junction Box (typically for one circuit)
JB1	Instrument Junction Box
EB	Equipment box (contains mainly electric equipment)
EC	Equipment cabinet (contains mainly electric equipment)
VS	Video Screen/Keyboard
PR	Printer
MC	Micro Computer
VB	Valve box (pneumatics)
HB	Valve box (hydraulics)
HP	Hydraulic panel
SVB	Solenoid valve box

AT	Automation Tower
HU	Hydraulic unit
LU	Circulation lubrication center
SU	Sump tank
LP	Lubrication panel

Example 1: 35-58CB-003
= Paper machine 3, pre-dryer section control box No. 003

Example 2: 35-24JBI-002
= Paper machine 3, broke collection, instrument field box No. 002

Example 3: 35-81CR-201.3
= Paper machine 3, winder 1, control cabinet 201, field 3

Example 4: DNA I/O cabinet field numbering
35-69CR-303.01F
= cable rise, left most field, front
35-69CR-303.02B
= cross connection, second field, back

Top view of the example cabinet [35-69CR-303](#)

(Back)

Cable rise 35-69CR-303.01B	I/O 35-69CR-303.02B	Cross connection 35-69CR-303.03B	Cable rise 35-69CR-303.04B	Equipment 35-69CR-303.05B
Cable rise 35-69CR-303.01F	I/O 35-69CR-303.02F	Cross connection 35-69CR-303.03F	Cable rise 35-69CR-303.04F	Equipment 35-69CR-303.05F

(Front)

2.1.3 Location Codes on field

TS	Tending side
DS	Drive side
FDG	Field
FDT	Field TS
FDD	Field DS
FRG	Frame
FRT	Frame TS
FRD	Frame DS

RS	Right side
LS	Left side

2.1.4 Circuit Position Codes for Hardware

MM-SSYYY-XXX

MM Plant Area code
SS Sub- Process code
YYY Circuit function code (1-3 characters)
Whenever possible only two letters will be used, example temperature control loop is TC, see examples.
XXX Circuit number (3 characters)
DCS and MCS, consecutive numbers will be used as follows:
001 ... 499 MCS (**EXCEPTION ! Winder MCS use 001 ... 999 area**)
500 ... 999 DCS
9000... 9249 local gauges MCS (loops)
9250... 9499 local gauges **DCS** (loops)

Example 1: 35-22PDI-165

Paper machine 3, short circulation/approach flow, differential pressure MCS circuit 165

Example 2: 35-57PC-436

PM3, press section, 3rd press roll air pressure, pressure control MCS circuit 436

Example 3: 35-85WC-201

PM3, winder 1, tension control MCS circuit 201

Circuit name: In documentation, max. 40 characters

2.1.5 The most common circuit function codes

Function	Circuit function code	Function	Circuit function code
Power control	EC	Level measurement	LI
Control voltage	EE ES	Level switch	LS
Flow control	FC	Moisture measurement	MI
Flow ratio control	FFC	Pressure alarm	PA
Flow measurement	FI	Pressure control	PC
Flow switch / alarms	FS	Pressure difference control	PDC

Start Up / Movement warning	GA	Loading pressure circuit (Rolls etc...)	PHZ
MCS movement control	GC	Pressure measurement	PI
MCS movement circuit	GHS	Pressure switch	PS
MCS movement circuit with interlocking	GHZ	Consistency control	NC Q€
Position / movement measurement	GI	Quality control measurement	QI
Limit switch	GS GO GS	Speed control	SC
Solenoid valve	SV GSV	Speed switch	SS
Movement interlocking	GZ	Temperature control	TC
Manually operated control valve	HC	Temperature measurement	TI
Manually operated valve indication	HI	HY/LUB center circuits	UIS
ON-OFF valve	HS	Stretching / force control	WC
Sheet break automatic	HSZ	Weight/force measurement	WI
Emergency stop	ES HZ	Vibration measurement	AI X†
Sequence control	KJ	Web break automatic	XS
Timer controlled function	KS	Interlockings	XZ
Level control	LC		

2.1.6 Circuit Position Codes for Software

MM-SSYYY-XXX-R

MM	Plant Area code
SS	Sub- Process code
YYY	Circuit function code (1-3 characters) Whenever possible only two letters will be used, example temperature control loop is TC, see examples.
XXX	Circuit number (3 characters) DCS and MCS, consecutive numbers will be used as follows: 001 ... 499 MCS (EXCEPTION ! Winder MCS use 001 ... 999 area) 500 ... 999 DCS 9000... 9249 local gauges MCS (loops) 9250... 9499 local gauges DCS (loops)
R	Consecutive number, if the circuit is divided into several program circuits (1-2 characters)

Example 1: 35-56HS-217
Paper machine 3, forming section, unit shifting MCS circuit No. 217

Example 2: 35-57PC-440-1
PM3, press section, 1st press loading pressure control, MCS circuit 440

Example 3: 35-57PC-440-3
PM3, press section, 1st press loading pressure control, MCS circuit 440

Example 4: 55-33FC-601
OCC production line, thickening, flow control in DCS circuit 601

Circuit name in the control room display: max 26 characters

2.1.7 Device Position Codes

MM-SSZZZNRR-XXX

MM	Plant Area code
SS	Sub-Process code
ZZZ	Component code (1-3 characters)
NN	Device serial number within the circuit, 1-99 (1-2 characters) (Note: Although if there is only one device in loop the serial number is always 1) , see examples below
R	Consecutive letter or number, if one device has several control signals (1-2 characters)
XXX	Circuit number (3 characters) DCS and MCS, consecutive numbers will be used as follows: 001 ... 499 MCS (EXCEPTION ! Winder MCS use 001 ... 999 area) 500 ... 999 DCS 9000... 9249 local gauges MCS (loops) 9250... 9499 local gauges DCS (loops))

Example 1: 35-56GSV1-217
PM1, forming section, solenoid valve No.1 in circuit 217
(more than 1 device in circuit)

Example 2: 35-56HS-217
= Control switch No.1 in circuit 217 (only 1 device in circuit)

Example 3: 35-56HS1A-217
= Control switch No.1, contact 1, in circuit 217
35-56HS1B-217
= Control switch No.1, contact 2, in circuit 217

For pneumatic solenoid coil 12 = A, coil 14=B

Example 4: 35-25HV1-717	Process valve
35-25GSV1-717	Pilot valve
35-25GS1O-717	Open limit
35-25GS1C-717	Close limit

In Software, I/O channels of the device must be named individually.

2.1.8 Automation Devices

Device	Symbol	Device	Symbol
Control switch	HS	E-stop pushbutton	ES HZ
Pushbutton	HS	Emergency switch	ES HZ
Illuminated pushbutton	HSI	Alarm siren	EA
Pilot light	GX	Photocell	GS
Panel meter, density	DI		
Panel meter, flow	FI	Flow switch	FS
Panel meter, position/length	GI	Proximity/Limit switch	GS GO GS
Panel meter, level	LI	Level switch	LS
Panel meter, pressure	PI	Pressure switch	PS
Panel meter, speed/frequency	SI		
Panel meter, temperature	TI	Temperature switch	TS
Panel meter, weight/force	WI		
Density sensor	DE	Density transmitter	DT
Flow sensor, -transmitter	FT		
Position/length sensor	GE	Position/length transmitter	GT
Level sensor	LE	Level transmitter	LT
Pressure sensor, -transmitter	PT		
Analysis sensor	QE	Analysis transmitter	QT
Consistency sensor	NE- QE	Consistency transmitter	NT QT
Speed/frequency sensor	SE	Frequency transmitter	ST
Temperature sensor (PT100)	TE	Temperature transmitter	TT
Load cell	WE	Load cell transmitter	WT
Optical trigger	ST	Vibration transmitter	AT VT
Break camera	BI	I/I-, I/U-, U/U-, U/I-transducer	TF
Socket inlet/outlet, Removable connector	EX	Frequency converter, Servo controller, Stepper motor controller	FC
Lamp	E H E	Main switch, Maintenance switch	S
Over voltage protector	FB	Power unit, Power transformer	T
Circuit breaker, Fuse	F	Optical isolator	U
Heater Heating resistor	H R	Stabilizer	V
Relays, contactors	K	LC filter, Diode	Z Ψ
Electric motor	M	Terminal block	X

Electrical brake	Q	Resistor	R
Main earthing busbar	MEB	Earthing busbar	EB
Protective ground	PE	Signal ground	FE
ProfiNet	PN		

2.1.9 Fluid Devices

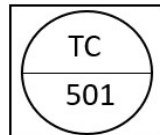
Device	Symbol	Device	Symbol
Hand valve	HV	I/P-converter	PCV
Pressure control valve, manual (e.g. pressure reducer)	PHV	Pressure transmitters	PT
Pressure proportional valve	PCV	Hose damage valve	PZV
Flow restrictor valve, manual	FHV	Ratio relay	PFV
Flow restrictor, fixed	FRV	Volume booster	PFV
Flow proportional valves	FCV	Hydraulic valve group	HVG
Shuttle valve	FVV	Pneumatic valve group	PVG
Check valve	FVV	Sub plate	SP
Quick exhaust valve	FZV	Cover plate	CP
Valve's open limit switch Valve's close limit switch	GSO G00- GS10 GSC G0C- GS1C	Measuring point	MP
Solenoid valve	SV GSV	Flow divider	FFV
Quick coupling	QCV		
Pump	P		
Hose	HSE		
Pressure gauge	PI		
Heat exchanger	E		
Accumulator / Container	C		
Filters	YU		
Vacuum Ejector	EJE		

2.1.10 Devices in PI-diagrams

Instrument loop position in PI-diagrams, square = DCS, diamond = MCS
In PI-diagrams the mill area and subprocess code is shown on the upper left of the instrument loop number box.

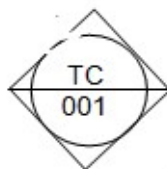
DCS-Loop

35-25



MCS-loop

35-25



2.2 AC-Motor, positioning and control principles

2.2.1 AC-Motor and MCC Positioning

The identifications of motor circuits are named according to equipment number.

Example 1, according to equipment number of

35-24P529	Process pump
35-24-529.1	Motor no. 1

Example 2, according to equipment number

35-24L101	Process equipment
35-24-101.1	Motor no. 1
35-24-101.2	Motor no. 2

Example 3, according to equipment number of

35-66L301	Hydraulic unit (equipment position)
35-66-301.1	Motor no. 1
35-66-301.2	Motor no. 2
35-66-301.3	Motor no. 3
35-66-301.4-R	Heater no. 1
35-66-301.5-R	Heater no. 2

2.2.2 Control Signals

Connection devices and signals:

Signals MCC - MCS x = In use (x) = In use when needed	In use	Device code	Signal code
Normal drive ND			
Control on/off (output)	x	-K1	CF
Run feedback on (input)	x	-K11	RF
MCC ready (input)	x	-K12	SR
Field fault (safety switch input)	x	-S11	FF
Reversing drive RD			
Control forward (output)	x	-K1	CF
Control reverse (output)	x	-K2	CR
Feedback forward (input)	x	-K21	RF
Feedback reverse (input)	x	-K22	RR
MCC ready (input)	x	-K12	SR
Field fault (safety switch input)	x	-S11	FF
Reversing drive two speed RD/2S			
Control forward speed 1 (output)	x	-K1	CF1
Control reverse speed 1 (output)	x	-K2	CR1
Control forward speed 2 (output)	x	-K3	CF2
Control reverse speed 2 (output)	x	-K4	CR2
Feedback forward speed 1 (input)	x	-K11	RF1
Feedback reverse speed 1 (input)	x	-K21	RR1
Feedback forward speed 2(input)	x	-K31	RF2
Feedback reverse speed 2 (input)	x	-K41	RR2
MCC ready (input)	x	-K12	SR
Field fault (safety switch input)	x	-S11	FF
Frequency converter SC			
Control on/off (output)	x	-K1	CF
Run feedback on (input)	x	-K11	RF
Speed control	x	-S11	SC
Rotation direction (forward/backward)	x	-R1	RD

Frequency converter ready	x	-K12	SR
Field fault (safety switch input)	x	-S11	FF
High Voltage motors			
Control on (output)	x	-K1	CF
Control off (output)	x	-K2	CS
Run feedback on (input)	x	-K11	RF
MCC ready	x	-K12	SR
Common signals			
Thermistor PCT (MCC input) for motors $P \geq \text{___ kW}$	(x)	-TE1/-TT1	TH
Winding temperature monitoring for motors $P > \text{___ kW}$ (AI signals will be connected to I/O-cabinet)	(x)	-TE1/-TT1	T1, T2, T3
Bearing temperature monitoring for motors $P > \text{___ kW}$ (AI signals will be connected to I/O-cabinet)	(x)	-TE1/-TT1	T4, T5
Current measurement (input) for motors $P > \text{___ kW}$ Measuring unit [%] (of I_n , [A])	x	-T1	CUR
Speed measurement (SI)	(x)	-T2	SI
Brake	(x)	-Q11	BR
Power (kW load)	(x)	-T3	LD
Diagnostic signals in bus control (max 10)	x		

Electric Device code

Device	Symbol	Device	Symbol
Electric motor	M	Main switch / Safety switch	S
Heater	H	Circuit breakers	F
Safety switch	S	Over voltage protector	F
Electrical Brake	Q	Measurement, current, power	T
Frequency converter, Servo controller, Stepper motor controller	FC	Lamp	E H E
Power units	T	Auxiliary relays, terminal block relay	K
Stabilizer	V	Temperature sensor	TE
LC filter	V	Temperature transmitter	TT

Example of signal and device positions

Position	Explanation
35-24P529	Process pump position code
35-24-529.1	Process pump, AC-motor
35-24-529.1	Motor circuit diagram
35-24-529.1	SW-circuit in control system
35-24-529.1-CF	Motor circuit diagram, Control On/off control relay tag (MCC I/O-cabinet)
35-24-529.1-CF	SW-circuit, control On/off, I/O tag in SW
35-24-529.1-K1	HW-circuit, Control On/off main relay in MCC
35-24-529.1-HS1A 35-24-529.1-HS1B	Special case... other devices in the AC-motor circuit Local switch, signal A Local switch, signal B

Example of winding and bearing signal and device positions

Position	Explanation
35-24-529.1T1 35-24-529.1T1 35-24-529.1-TE1 35-24-529.1-TT1	Circuit code for winding 1 temperature DCS AI, L1 winding signal tag L1 winding temperature sensor L1 winding temperature transmitter
35-24-529.1T2 35-24-529.1T2 35-24-529.1-TE2 35-24-529.1-TT2	Circuit code for winding 2 temperature DCS AI, L2 winding signal tag L2 winding temperature sensor L2 winding temperature transmitter

35-24-529.1T3 35-24-529.1T3 35-24-529.1-TE3 35-24-529.1-TT3	Circuit code for winding 3 temperature DCS AI, L3 winding signal tag L3 winding temperature sensor L3 winding temperature transmitter
35-24-529.1T4 35-24-529.1T4 35-24-529.1-TE4 35-24-529.1-TT4	Circuit code for shaft side B1 bearing temperature DCS AI, shaft side B1 bearing temperature Shaft side B1 bearing temperature sensor Shaft side B1 bearing temperature transmitter
35-24-529.1T5 35-24-529.1T5 35-24-529.1-TE5 35-24-529.1-TT5	Circuit code for non shaft side B2 bearing temperature DCS AI, non shaft side B2 bearing temperature Non shaft side B2 bearing temperature sensor Non shaft side B2 bearing temperature transmitter

2.3 Positioning, QCS

Modules and process measurements concerning quality control are named according to QCS naming convention.
QCS naming convention is presented in below.

2.3.1 Primary designations

MID	Machine identifier
LID	Profile name, that consists of:
XX	Profile type acronym
POS	Scanner location acronym

In mapping the actuator to scanner location the naming is:

ACT	CD actuator acronym
POS	Scanner location acronym

2.3.2 Machine ID (MID)

Machine IDs always followed by a dash (-).

BMn	Board Machine n
CLn	Soft Calender n
CMn	Coating Machine n
CTn	Standalone Coater n
GCn	Gloss Calander n
PDn	Pulp Dryer n
PMn	Paper Machine n
Pnn	Paper Machine nn
SCn	Supercalander n
TMn	Tissue Machine n

n= 1, 2, ...

2.3.3 Profile Type Acronyms (XX)

BW	Basis Weight
----	--------------

OD	Oven Dry Weight
NW	Normalized (conditioned) Weight
WW	Water Weight
PM	Percent Moisture
PD	Percent Air Dry
CW	Coat Weight
SW	Size Weight
TA	Total Ash Content
AP	Ash Percentage
GL	Gloss
BR	Brightness
WH	Whiteness
OP	Opacity
FR	Formation
SM	Smoothness
CA	Caliper
RH	Reel Hardness

2.3.4 Scanner Location Acronyms (POS)

RL	at Reel
BL	at Baler
CL	at Calander
Ln	after Calander n
L0	before Calanders
Cn	after Coater n
C0	before Coaters
Dn	after Dryer section n
D0	before Dryers
YK	after Yankee
Yn	after Yankee n
SP	at Size Press
Sn	after Size Press n
S0	before Size Presses
WP	after Wet Press
PU	at Pickup
Pn	at Pickup n
Fn	above Former n

n= 1, 2, ...

2.3.5 Scanner Difference Acronyms

nC	difference between Coater n and Coater n-1
nD	difference between Dryer n and Dryer n-1

2.3.6 Profile Name Examples

PM1-BWSP	Basis Weight at Size Press
PM1-BWRL	Basis Weight at Reel
PM1-PMSP	Percent Moisture at Size Press
PM1-ODSP	Oven Dry Weight at Size Press
PM1-CWC1	Coat Weight after Coater 1

2.3.7 Control variables

Quality variable controls follow profiles naming convention.

Master level controls:

ODFF	OvenDryFeedForward weight (if only one headbox)
ODFT	Total feed forward weight (in multistock application)
ODF1	OvenDryFeedforward for Headbox 1
ODF2	OvenDryFeedforward for Headbox 2
ODFn	OvenDryFeedforward for Headbox n
JWC1	Jet/Wire Control for Headbox 1
JWC2	Jet/Wire Control for Headbox 2
JWCn	Jet/Wire Control for Headbox n
TAFF	TotalAshFeedForward
CSC1	Coordinated Speed Change
MSO1	MachineSpeedOptimization
DLC1	DryerLimitedControl
IGC1	AutomaticGradeChange
APL1	AutomaticPressureLetdown
MXM1	MaXimumMoisture

Low level controls:

TSF1	ThickStockFlow
S1SC	Section 1 Steam Control
HBTH	HeadBox TotalHead (if only one headbox)
HBT1	HeadBox Totalhead for Headbox 1
HBT2	HeadBox Totalhead for Headbox 2
MSC1	MachineSpeedControl
FIF1	Filler Flow

Process measurements:

TSC1	ThickStockConsistency
WRS1	WiReSpeed
RLS1	ReeLSpeed
SLO1	SLiceOpening
RLD1	ReeLDiameter
LCRT	CIE L Top
ACRT	CIE a Top
BCRT	CIE b Top

OPRT	Opacity Top
LCRB	CIE L Bottom
ACRB	CIE a Bottom
BCRB	CIE b Bottom
OPRB	Opacity Bottom

2.4 Positioning, lubrication and condition monitoring

2.4.1 Device position codes for lubrication

MM-SSACCC-FX

MM	Plant Area code
SS	Sub-Process code
A	Basic equipment identification code
CCC	Consecutive number for unique equipment
FX	Lubrication point code

Lubrication point (FX)

F1	Bearing TS
F2	Bearing DS
F3, F4	Roll gear
F5, F6	Gear
F7, F9	Bearing for intermediate shaft, machine side
F8, F10	Bearing for intermediate shaft, motor side

Example 1 35-56ACCC-F1
PM1, forming section, roll no. CCC , bearing lubrication TS

2.4.2 Device position codes for condition monitoring

MM-SSACCC-FX-VTX

M	Plant Area code
SS	Sub-Process code
A	Basic equipment identification code
CCC	Consecutive number for unique equipment
VT	Component code
X	Device serial number within the circuit

Conditioning monitoring point component code (VT)

VT	Vibration transmitter
VT1	Bearing TS
VT2	Bearing DS
VT3, VT4	Roll gear
VT5, VT6	Gear
VT7, VT9	Bearing for intermediate shaft, machine side
VT8, VT10	Bearing for intermediate shaft, motor side
PT	Pressure transmitter
SN	Roll trigger (Magn.)
ST	Felt trigger (Optc.)
STL	Felt trigger (Light)

Example 1 35-56ACCC-VT1
PM1, forming section, roll no. CCC , vibration measurement
no. 1 (Bearing TS)

2.5 Terminal blocks and Internal wiring

2.5.1 Terminal block numbers

The series numbers for terminal blocks are as follows:

X1/1, 2, 3...
X2/1, 2, 3...
|
X999/1, 2, 3...

X1 and X2 are terminal strip identification symbols where 1, 2, 3... are terminal numbers.

2.5.2 Internal wiring color codes in drawings

The colors are based on the standard IEC 60757

Color	Code
Black	BK
Brown	BN
Red	RD
Orange	OG
Yellow	YE
Green	GN
Blue (including light blue)	BU
Violet (purple)	VT
Grey	GY
White	WH
Pink	PK
Gold	GD

Turquoise	TQ
Silver	SR
Green-and-yellow	GNYE
Cable shield	SHD
Pair shield	TE1, TE2, ...

2.5.3 Wire Colors, Internal Wiring

Colors are based on the standard IEC 60204

Voltage level	Comment	Color
12 - 48 VDC	Positive (+) and negative (-)	Blue
12 - 48 VDC	Wrap wire	Black
12 - 48 VAC	If both phases are galvanically isolated	Red
12 - 48 VAC	If one is grounded	Blue (light)
12 - 48 VAC	Phase	Red
230 VAC	Phase (L1)	BROWN Black CHECK IEC 60204 !
	0 (N)	Blue (light) Different from TE and 12-48 VDC (Row1) CHECK IEC 60204 !
	(PE)	Yellow-green CHECK IEC 60204 !
	(TE)	Blue
110 VAC	Phase (L1)	Red
	0 (N)	Red
	(PE)	Yellow-green
	(TE)	Blue
Foreign voltage	Interlocking, supplied by external voltage supply	Orange

2.5.4 Wire Marks, Internal Wiring

Cabinet and boxes internal wiring with wire marks
Cabinet wrap wiring without wire marks

2.5.5 Cabinet's and Box's Internal Wiring

Voltage	Min. cross-sectional area
---------	---------------------------

24 VDC and AC	0.5 mm2 (wrap wiring 0,2 mm2)
230 VAC	1.5 mm2

2.6 Colors of push buttons and pilot lights

The colors are based on the IEC standard IEC 60073

Pushbuttons	
Emergency Stop	Red
Stop/Off	White
Start/On	White
Reset	White
Quick Stop	Black

Illuminated pushbuttons	
Stop/Off - Start/On (Status, mode, etc.)	White/clear

Pilot lights	
Motor runs	White/clear
Control device in certain position, e.g. on the limit	White/clear
Control device in safe/normal state, e.g. in driving position or interlocking doesn't prevent device from moving, in other words movement is allowed	Green
Control device in abnormal state, e.g. interlocking prevents movement	Yellow
Special meaning, state that requires actions, e.g. break state indication	Blue

The movement of the control device towards the controlled position is indicated by a flashing signal light.
Two different frequencies are in use; $f_1 = 0,5 \text{ Hz}$ and $f_2 = 2 \text{ Hz}$. If only one frequency is in use, the frequency is $f_2 = 2 \text{ Hz}$.

2.7 Cable numbers and principles

2.7.1 Cable Numbers

The cable numbers are based on the IEC standard IEC 81346-2

Instrument cables

Multicore cables

Multicore cables between the DCS system and junction box, PLC system, panels, etc. shall bear the number of the “satellite” accompanied by a consecutive number separated by a dash.

New example:

35-22JBI-033-1
Instrument junction box 033, area 35,
sub process code 22
with multicore cable no. 1

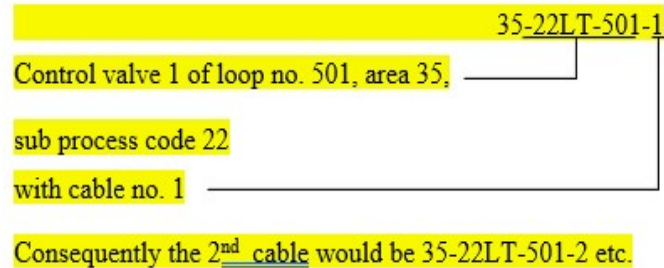
Consequently the 2nd multicore cable would be 35-22JBI-033-2 etc.

35-22CP-221-1
Control panel 221, area 35,
sub process code 22
with multicore cable no. 1

Local cables

Local cables from field instruments and similar to junction boxes, PLC systems, panels, etc. shall bear the device tag accompanied by a consecutive number separated by a dash.

New example:



~~Location code WXN or device position code WXN~~

Location code /	Cabinet or box code from where the cable is coming or
Device position code	device position from where the cable is coming
WX	Cable code
N	serial number

CHECK BELOW !

Cable codes

Cable codes (-WXXX)	Cable function
-WE1 ...	Automation grounding cables
-WC1 ...	Automation power supply cables
-WF1 ...	Automation signal cables
-WF101 ...	Automation system cables system cables

Examples of cable numbering (rev 05, 26.10.2022)



Cablenum_eng2019 rev5.pptx

~~Location code – WXN or Device position code – WXN~~

Location code	cabinet or box code from where the cable is coming see item 2.2.2
Device position code	device position from where the cable is coming see item 2.2.6
W	cable identifier
N	serial number

~~Serial numbers are divided into groups according to cable function as follows:~~

~~WC1...WC4: Feeder cables without separate frequency converter~~

Cable codes – (WX)	Cable function
WC1	MCC – Safety switch If used parallel cables: W1A, W1B...
WC2	Safety switch – Motor
WC3	MCC – Safety switch (2-speed)
WC4	Safety switch – Motor (2-speed)

~~WC1...WC3: Feeder cables with separate frequency converter~~

Cable codes –	Cable function
WC1	MCC – Frequency Converter If used parallel cables: W1A, W1B...
WC2	Frequency Converter – Safety switch
WC3	Safety switch – Motor

~~WC5... WC10: Not specified~~

~~WG11...WG20: Standardized control cables~~

Cable codes –	Cable function
WG11	MCC – Control system (bin.)
WG12	MCC – Safety switch
WG13	MCC – Motor thermistor
WG14	MCC – Control system (analog)
WG15...WG20	Not specified

~~Wx21... Wx99: Other control cables for MCC
(x is the subclass of cable, based on kind of processing)~~

Motor cables

Example: 25-22P529 Mechanical position

25-22-529.001-W1

= Pump No. 529, motor No. 1, Main cable from MCC to the motor

BELOW FROM ELECTRICAL CABLES FROM STANDARD PS-21 !

W1	Power cable.
W2	Power cable.
W3	Power cable.
W4	Power cable.
W5	MCC - Control Rack.
W6	Current 0 ... 1A.
W8	Not yet specified.
W9	Not yet specified.
W10	Not yet specified.
W11..	All other control cables

2.7.2 Cabling Principles

Field cables are connected in number/color code order to cable-specific terminal strips.

Cabling of paired cables starts with the first conductor wire of the pair.

24 V analog and binary signals are hardwired to ~~the same-different~~ cables
~~Analog trunk cables are 'Jamak' type~~, and ~~binary trunk cables are~~
~~'Nomak' type~~

Trunk cable spare conductor wires are connected to terminals

2.8 Field Cable Types

INFORMATION NEEDED FROM SHOTTON MILL !



Appendix_3_cable types.xlsx

2.9 Hydraulic, pneumatic and lubrication piping

2.9.1 Small bore piping numbers for hydraulic and pneumatic.

MM-SSYYY-XXX-EENN

MM	Plant Area code
SS	Sub-Process code
YYY	Circuit function code
XXX	Circuit number
EE	Application area
NN	Running number inside one circuit (1-2 characters)

Application area

Hydraulic pipe, pressure line	P
Hydraulic pipe, tank line	T
Hydraulic pipe, drain line	DR
Hydraulic pipe, actuator line	H
Pneumatic pipe	PP

Example 1: 35-56GHZ-217-T1
Hydraulic pipe, tank line in circuit 35-56GHZ-217

Example 2: 35-56PC-207-PP2
Pneumatic pipe in circuit 35-56PC-207

Pipe codes are marked onto all pipes leaving the centers and valve panels.

2.9.2 Small bore piping numbers for lubrication

MM-SSACCC-FX

MM	Plant Area code
SS	Sub-Process code
A	Basic equipment identification code
CCC	Consecutive number for unique equipment
FX	Lubrication point code

Lubrication pipe, pressure line	Lubrication point position
Lubrication pipe, tank line	TL

Example 1 35-56ACCC-F1
PM1, forming section, roll no.CCC , bearing lubrication TS

2.9.3 Pipe Standards

Pipings are made according to Piping Standards for Small-Diameter Pipes (in Technical Specification)

2.10 Measuring units

~~TO BE CHECKED!~~

OK, accepted by the Customer on May 18, 2022

Variable		US
Pressure	Pa, kPa, MPa	PSI
Nip load	kN/m	PLI
Temperature	°C (Celcius)	°F (Fahrenheit)
Tension	N/m, kN/m	PLI
Speed	m/min	fpm
Power	kW	hp
Voltage	V	V
Current	A	A
Rotation speed	r/min	r/min
Dimensions	mm, m	inch, ft
Flow	m3/s, l/min, l/s	
Mass flow	kg/min, kg/s	
Consistency	%	
Density	kg/dm3	
Mass	g, kg	
Basis weight	g/m2	
Concentration	pH	
Viscosity	mpas (Brookfield)	-

Dissolved oxygen	0 - 15%	
Level measurement	0-100%	0-100%
Valve position	0-100%	0-100%
Motor current meas.	0-150% (100%=In)	0-150% (100%=In)
Sectional drives torque	0-100% (100% = RDC at max. machine speed)	

2.11 I/O Interface Methods, ValmetDNA

2.11.1 Connection method

- 24 VDC cross connection in I/O cabinets will be done with **screw terminals**
- Connection in other cabinets, boxes and panels will be done with screw terminals

2.11.2 I/O-Card types

- Without Hart connection
- Centralized I/O-cards are **without** protective coating (laquered)
- Distributed I/O-cards are threaded with a protective coating (laquered)
- Binary Input and output cards' are Positive, PNP-logic

- Recommended I/O-units are as follows:

ValmetDNA, M80 Series	
Analog units (AI/AO)	
<u>AI8C</u>	- 8 analog input channels - 0/4...20 mA
<u>AO4C</u>	- 4 analog output channels - 0/4...20 mA
Digital units (DI/DO)	
<u>DI8P</u>	- 8 digital input channels - self-powered, short-circuit protected, opto-isolated, PNP input (24 VDC)
<u>DO8P</u>	- 8 digital output channels - self-powered relay output 200 mA/24 VDC

The letter L at the end of the product ID indicates a lacquered product.

2.12 Grounding principles, ValmetDNA



Link to Valmet grounding principles [Grounding of metsoDNA automat.pdf](#)

Copy the document from master document, Grounding of ValmetDNA 

2.13 Process Engineering Instructions

2.13.1 Process piping

SEE SHOTTON PROJECT STANDARD PS-21 !

2.13.2.2 Abbreviations for flow substances and raw materials

SEE SHOTTON PROJECT STANDARD PS-21 !

2.13.3 Pipe material codes

SEE SHOTTON PROJECT STANDARD PS-21 !

2.13.4 Pipe pressure classes

SEE SHOTTON PROJECT STANDARD PS-21 !

2.13.5 Process area hand valves

SEE SHOTTON PROJECT STANDARD PS-21 !

2.13.6 Valve type abbreviations and codes

SEE SHOTTON PROJECT STANDARD PS-21 !

2.14 General control principles and definitions of process automation (DCS)

2.14.1 General

2.14.1.1 Location of operating controls

The mill has control room for every main area. The process is operated and controlled via DCS operator workstations located in the control rooms. Operator workstations include necessary monitors, computer keyboard, and mouse systems with necessary printers for alarm and report output devices.

There are the local control boxes and control panels for some process equipment for maintenance and limited local control purposes. The control boxes are located always near the process equipment. Local controls are connected to DCS system.

2.14.1.2 Distributed Control System (DCS)

The process control system controls the process instruments and motors according commands given by operator and defined automatic functions. The process control system is a new generation DCS, using open system technology to provide flexibility and be capable of supporting at the top of its basic control level, the implementation of supervisory controls, automatic start and stop sequencing, communication interface with other systems (via Profibus, Modbus or OPC interfaces), always trying to use the “single-window” concept for operation interface. The communication interface with Motor Control Centers is highly reliable and effective in order to enable exchanging command, status, diagnosis and interlock signals.

There are separate engineering stations for each main area with “on-line” programming and partial/total “download” capability, “force/trace” resources associated with capability for “on-line” viewing of signal and interlock status. The system has also self-documentation tools, software testing tools and control simulation tool without the need for physical I/O cards.

2.14.1.3 Operating controls

The system includes efficient man-machine interfacing through operator workstations, with dual video display arrangements, in order to allow a larger number of information available per operator.

Process is operated via graphic displays opened on the operator workstations. Graphic displays consist of a basic picture and active points defined for the display. Active points are used to output measurement and other value information, graphic valve and motor symbols, various data entry fields and points to call up detail displays and loop pictures on the display. For detailed study, tuning and control of all basic loop functions the system include detail loop pictures of those functions. For process monitoring there are available necessary trend displays and history database.

2.14.2 TAG NUMBERING SYSTEM

All instrument and motor loops are individually tag numbered according Valmet tag numbering standard. All program loops, sequences and group controls etc. have also own individual tag numbers.

2.14.3 PROCESS CONTROL PRINCIPLES

The process description comprises the main operating principles of the process and a general description of the group starts and interlock systems. Detail information of controls, logic and interlocks are presented in the loop functional diagrams.

Basic process control functions are presented in this document in order to simplify and clarify process descriptions. Basic principles for typical and most common control cases are explained. These principles are normally applied in all process areas. If exceptions are needed they will be mentioned case by case in process descriptions.

Detailed instructions for the machines, such as the paper machine process equipment, pulpers, screens, dispersers, refiners and cleaners, etc., are given in the documentation and operating manuals provided with the machinery.

2.14.4 BASIC CONCEPTS

2.14.4.1 Operation modes

MANUAL mode

In MANUAL mode device is controlled by operator if not prevented by an interlock or forced control. For example motor can be started or stopped, on/off-valve can be opened or closed, controller output can be adjusted (no automatic control actions are performed).

AUTO mode

In AUTO mode motors, on/off-valves and other binary functions are operated by a group controls or other logic functions. Controllers perform their control actions according to active set points, measurement values and tuning parameters.

Set point selection

Controller's active set point will be selected by set point selection mode. In LOCAL AUTO mode controller follows the LOCAL set point adjusted by operator. In REMOTE AUTO mode the set point will be calculated or controlled by remote function according process description. In REMOTE AUTO mode LOCAL set point follows REMOTE set point to allow smooth and bumpless mode change. In computer mode (special case of REMOTE mode) the active set point will be determined by upper level control system. Normally upper level control system is quality control system.

LOCAL control

If motor is equipped with local controls and is selected to LOCAL mode, motor is not operable from display and automatic controls are ineffective. In LOCAL mode motor is controlled via DCS. Local control switches are always connected to DCS and are intended to use for maintenance and limited local control purposes. Local control is selected and operated from switches located to local control box. In LOCAL mode there can be less interlocks than in MANUAL and AUTO mode. Switching the mode back to REMOTE motor will stay stopped, until the next DCS start command.

Tracking

When loop mode is changed output value is usually tracked to same value as it was in previous mode. For example controller output value remains same when mode is changed from AUTO to MANUAL. When motor loop mode is changed motor running condition remains same if not stopped by an interlock.

2.14.4.2 Group controls (group Start sequences)

General

To help process start-ups and shut-downs the motors and instrument loops of same sub process areas have been clustered in to groups. The purpose of group Start and Stop controls is to avoid mistakes in the operation and to have a faster start-up and shut down. The group members are collected in to a group Start display by starting order. Display can be activated when necessary. A group can be started and stopped with one command from the operator workstations. Devices belonging to a group can be disconnected from the group and be controlled individually all the time via operator workstations if necessary.

Group Start and Stop commands affects only to motors selected to AUTO mode. Group Start and Stop skips motors selected to MANUAL mode. Operator may select motors to MANUAL to keep them stopped after group Start and to keep them running after group stop. Interlocks are effective all the time.

Group control information's, sequence step names and step transition conditions will be presented in the group or sequence faceplate with clear names.

MANUAL operation mode of sequences is allowed. There are applications that require ability to proceed in sequence step by step in special situations.

Group mode change command

Separate mode change commands included to group starts change modes of all necessary loops belonging to group if the loops are not already in correct mode.

All loops can remain usually in highest mode all the time if process is in normal state. Operator may change loop modes and operate individual loops after group controls all the time when necessary. Interlocks do not either change operation modes of the loops.

Group Start sequence

Main principle is that Group Start sequences will operate loops only as a pulsed command.

Group Start sequences are designed usually for starting process from condition when main tanks are already filled. For example automatic water fillings for pressure screens and some other process equipments are included in group Start sequences. When process is started from empty condition always some MANUAL operations are needed.

Group Start sequences operate loops only as a pulsed command. Group Start sequences proceeds step by step with necessary predefined delays according specified order. The correlations between process functions define the starting order of the equipment. Starting order is defined so that previous steps usually clear interlocks for later steps. Some interlocks are temporarily cleared for start up by separate logic and will be effective after delay during start up

Each step in a sequence has own time monitoring and alarm will be given if the monitoring time is overrun.

Group Stop sequence

Main principle is that Group Stop sequences will operate loops only as a pulsed command.

Group Stop sequences are designed usually for short shut downs. Automatic water flushing for some process equipments are included in groups stop sequences. When process is stopped for long shut down period, some MANUAL operated flushing and tank emptying operations etc. are always needed. Group Stop sequences do not change operation modes.

Group Stop sequences operate loops only as a pulsed command. Group Stop sequences proceeds step by step with necessary predefined delays according specified order. The correlations between process functions define the

stopping order of the equipment.

Parallel and stand-by process devices

If the process line has a parallel and stand-by process devices, a selector loop will be used for selection. Bypass selector can be accomplished with software Bypass selector switch on DCS display or by hand valves equipped with limit switches or by automatic on/off valves. Manner how by pass selector loop is accomplished depends of application how often bypass selection has to be made and also safety issues. Group control commands to loops of bypassed process areas are prevented. Interlocks from loops in bypassed process areas are ineffective to other loops.

Sequence resetting

If group control or sequence sets continuous run commands or other continuous control commands to individual loops, each control command is reset when sequence is reset.

2.14.4.3 Interlock principles

General

To minimize the effect of a process disturbance or a device failure most of the instrument and motor loops have been interlocked with each other or with another process device. An interlock is a function which protects personnel from injuries and protects a machine or device against breaking or other damage by setting the device control to a safety state. Normally this state is stopped or closed depending on the device. The interlocks have two functional principles:

- Absolute type interlock set the control to safe state and prevents the control all the time in all modes, as the interlock is valid. Interlock is continuously effective in all modes. Absolute type interlocks are usually used for motor controls and safety related instrument loops.
- Pulse type interlock sets the control to safe state with a pulse. After the pulse the device is operable again in MANUAL mode. In AUTO mode interlock is continuously effective. Pulse type interlocks are usually used for normal instrument loops.

Conditions of individual Interlocks will be presented in the loop faceplate with clear signal names.

Interlock by pass is allowed for operator only for some selected motor and instrument loops according descriptions. Safety related interlocks cannot be by passed.

Each equipment and Group Start/Stop sequence should have own Interlock Window where can be seen

- Status of each individual interlock
- Fixed interlock value for analog measurements and calculated values
- Actual process value of the measurement or calculated value

- A sign to show the first interlock signal which caused the last interlock of this equipment or Group Start/Stop sequence
- The sign must not be removed before next successful start/control of this interlocked equipment

Motors

Interlocks for motor loops are usually continuously effective in all modes. Only some specific low importance process interlocks are bypassed in MANUAL and LOCAL mode by separate logic according descriptions.

Usually motors do not start automatically after interlocks. Group start or MANUAL start is needed for starting motor. Only some of the motors in special cases are allowed to restart in AUTO mode after some selected interlocks, issue is always mentioned in descriptions.

Some process interlocks only prevent starting of the motor and do not affect if motor is already running (start interlock).

Interlocks do not change operation modes of the motor loops.

Equipment motors

Main principle is that Interlocks for equipment motors are continuously effective in all modes.

Some special equipment motors have automatic functions and motors are allowed to restart in AUTO mode automatically after some selected interlocks. Some MANUAL and LOCAL operations are allowed by separate logic according descriptions. Equipment motor detail functions are always done according application.

Process pumps

Interlocks for process pump motors are continuously effective in all modes except feed tank low level and receiving tank high level interlocks which are bypassed in MANUAL mode by separate logic according descriptions.

Tank low level interlock is bypassed for normal pumps in MANUAL mode by separate logic for specified time. If feed tank level is below low low more than specified time pump will stop with a pulse, pump can be re started manually and pump can run only specified time if level remains below low low. Receiving tank high level interlocks are bypassed in MANUAL mode. In AUTO mode interlocks are continuously effective without delays.

Some pump motors are allowed to restart in AUTO mode after some selected interlocks if mentioned in descriptions.

Agitators

Agitators are interlocked in all modes to low level of the tank. When low level stops agitator, only low level alarm will be displayed. Agitator will start in AUTO mode when level is above low with defined hysteresis. MANUAL operation is not available if the tank level is below low level. There are usually no interlocks

from agitators to other loops

Motor protection

Some of the motors have winding and bearing temperature sensors with alarm and interlock signals to protect motor from overheating. Motor high (H) temperature alarm signal will only alarm. Motor high (HH) temperature interlock signal will alarm and stop motor (if information or signal is available).

Speed monitoring switches

Equipment speed monitoring switches will interlock only motor of monitored equipment. There are necessary starting delays in interlocks to allow start up. Interlocks for other loops are usually connected to motor control loop not to speed monitoring loop.

Instrument loops

Main principle is that Instrument loop forced controls and interlocks are usually continuously effective only in AUTO mode, MANUAL mode forced controls and interlocks are pulsed commands.

Permanent interlocks in MANUAL mode are used only when necessarily needed. Basically only most important safety interlocks are valid continuously in all modes. Interlocks will control instrument loops in MANUAL mode usually as a pulsed command. MANUAL mode pulsed interlocks allows operator to operate valves for flushing and filling purposes when process is not running. In case of a controller and control valve with hand station interlock (forced control) means that output value for device is forced to certain predefined value which usually is 0 %. When control loop mode is changed the loop output value must be tracked to prevent unnecessary functions. When device is interlocked in MANUAL mode device should remain its interlocked position after interlock is released.

When controller is released from forced control, the output follows smoothly the I-factor of the controller to new control value, without any jump to new output value. If any fixed ramps after releasing controller from forced control are needed, they have been specified separately.

In interlock cases devices should go usually fast to interlocked position. Specific ramp functions are usually used for startup, shut down and anti clogging etc. functions like described in process descriptions and function diagrams.

Interlocks do not change operation modes of the instrument loops.

Startup valves

Startup valves are usually allowed to operate manually even if the pump is not running. Startup valves have to be closed always when pump starts. In AUTO mode startup valves are slowly opened automatically after pump starts. If the startup valve does not open during predefined time pump will stop. Closing of the start up valve when pump is running will stop the pump after delay. Some specific pump start valves should close slowly to avoid vacuum in pipe line, such cases pump runs until valve is closed.

Pump suction side on-off valves

Pump suction side on-off valves are set to AUTO-mode and opened by the group start before pump is started. These valves will be closed by a pulse interlock when the pump stops.

Sealing water

Generally equipments with mechanical seals are equipped with sealing water flow monitoring. Sealing water low flow signal prevents equipment to start. There are usually 5 s delays in sealing water low flow signals to prevent unnecessary alarms, if signal is used for direct interlock then 2 s delay is used. Low flow signal will stop equipment usually after predefined delay. For example there are no seal water interlocks for under machine pulpers. For machine screens there is only start interlock.

There are also sealing water pressure and pressure difference interlocks for some specified process equipment, for example refiners and dispersers.

Water valves

Sealing water valves are prevented to close if any devices are running or are in stand by condition in sealing water group. Sealing water valves will close if sealing water pumps stops.

Dilution water, sealing water, consistency control and other water valves are always interlocked according to loop interlock type (pulsed or permanent) with other interlocks to line's water feed pump.

Measurement forcing to zero

Flow, consistency, quality and analysis measuring values shown in the graphic displays are forced to zero value when the sub process's is not running. Measurement zeroing signal will be taken always from pump which feeds process substance to measuring point.

2.14.4.4 Control principles and initial parameters

Basic task of the controller is to control the process variables according to control directions designed for the application. The controller input value is typically process measurement value from a transmitter. The output values are controlled in AUTO mode according measurement, active set point and tuning parameters. Controller has necessary MANUAL mode operations available and also interlocks and forced controls.

If controller has more than one output device each output will be done with separate HI- valve controller block. When main controller is selected to AUTO mode each valve controller is pulsed to AUTO and REMOTE mode. Each output device has individual interlocks.

In motor control loop it is not possible to set the speed set point

(AUTO/MANUAL) for frequency converter. Speed set points (AUTO/MANUAL) will come always from the corresponding instrumentation control loops.

If cascade control master controller is switched to AUTO mode, slave controllers will be switched to AUTO and REMOTE mode as a pulse. If master controller is in AUTO mode and have a measurement failure, slave controller is forced to LOCAL mode.

Controller maximum output rate of change (effective in all modes and control and interlock cases) and maximum set point rate of change should be possible to be limited.

All control valves and other analog controlled valves must have pre-defined opening and closing times in DCS software ($x\%$ / second), own values for MAN, AUTO and FORCE CONTROL-modes.

This is to help process controls, pumps, pipelines etc when special slow or fast opening / closing times are needed.

Following initial values for the parameters of controllers (K_p / T_i) will be used:

Flow	0,2/10
Level	1,5/200
Pressure	0,5/15
Consistency	0,5/90 DNA measurement filter 9,1s
Temperature	1/150
UTM Pulper levels	4/60

2.14.5 PROCESS ALARMS

Alarms are generated according process demands. An alarm will be issued when defined binary value is in a wrong state or when an analog value is beyond predefined limits. Alarm limits are described in interlock and alarm lists.

Alarms will have necessary hysteresis to avoid repeated unnecessary alarms when process variables are near alarm limit. Unnecessary alarms must be also prevented (masked) during start up and when process is not running. Flow, consistency, quality, analysis and low-pressure alarms are not effective if the sub process's feed pump is not running. There are necessary starting delays before low alarms will be active to allow alarm free start up. All interlocks are valid all the time even if alarm is prevented. Motor function blocks must have internal delay to prevent peak current alarm when starting motor.

2.14.6 FAULT SITUATION

In fault situations motor is stopped and mode is forced to MANUAL. Fault alarm will be displayed. Motor is operable by the operator.

In measurement fault situations controller mode is forced to MANUAL and output is freezed to its previous value. Measurement fault will give an alarm. Controller output is operable by the operator.

2.14.7 OTHER INFORMATION

The function description is consultative as regards accessories and equipment delivered by other suppliers. Necessary information of controls, logic and interlocks for machines supplied by other supplier than Valmet Paper has to be delivered by each equipment supplier.